

Project Title: Robust Cross-Domain Sentiment Analysis with BERT: PEFT vs Full Fine-Tuning

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Problem, Motivation, and Project Scope

This project studies sentiment analysis under domain shift, where models trained on one domain (e.g., movie reviews) often perform poorly on other domains such as product and restaurant reviews. This is a common issue in real-world NLP applications where data distributions vary.

We use BERT as a baseline model and implement the standard fine-tuning approach for SST-2 as described in Devlin et al. (2019). The main focus of the project is to compare different fine-tuning strategies, including full fine-tuning and parameter-efficient methods (such as LoRA and prompt tuning).

We evaluate these methods in terms of cross-domain generalization, robustness to input variations, confidence calibration, and computational efficiency. The motivation is to understand how modern NLP models can be adapted efficiently while maintaining performance in realistic and changing data conditions.

Seminal Paper, Dataset, and GitHub Reference

The project is motivated by the **seminal paper**: Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*.

We implement the standard BERT fine-tuning pipeline for sentiment classification and use it as a baseline for comparison.

Datasets:

- SST-2 (in-domain training and evaluation)
- Yelp Reviews, IMDB Reviews, and Amazon Reviews (cross-domain evaluation)

Reference GitHub Repository: Hugging Face Transformers – text classification examples python files (e.g., `run_glue.py`) used as reference for implementing the training and evaluation pipeline.